QudoSSL Commercial — Performance (Linux aarch64)

Unmodified upstream **OpenSSL 3.5.7** — the QudoSSL Commercial product (no Community/delegated build involved). Measured with OpenSSL's own `speed` and `perftools` handshake tools, median of 6 repetitions, single core. Higher is better.

Environment

CPU	Apple M4 (Linux arm64 container)
Architecture	aarch64
SIMD	neon
Cores	10
OS	linux
Kernel	Linux 6.12.76-linuxkit
OpenSSL	OpenSSL 3.5.7 9 Jun 2026 (Library: OpenSSL 3.5.7 9 Jun 2026)

Primitives — FIPS provider (`openssl speed`, operations / sec)

Key encapsulation — ML-KEM and hybrid groups

Group	keygen	encapsulate	decapsulate
ML-KEM-512	40,175	177,007	109,199
ML-KEM-768	27,592	132,512	80,420
ML-KEM-1024	19,652	99,249	61,662
X25519MLKEM768	20,399	25,966	33,176
SecP256r1MLKEM768	10,549	10,520	23,816
SecP384r1MLKEM1024	532	455	2,533

Signatures — ML-DSA

Algorithm	keygen	sign	verify
ML-DSA-44	2,355	3,304	17,600
ML-DSA-65	1,486	2,003	11,447
ML-DSA-87	1,177	1,752	7,338

Primitives — default provider (`openssl speed`, operations / sec)

Key encapsulation — ML-KEM and hybrid groups

Group	keygen	encapsulate	decapsulate
ML-KEM-512	100,365	177,746	109,698
ML-KEM-768	62,290	131,307	79,934
ML-KEM-1024	40,737	100,080	61,510
X25519MLKEM768	33,270	25,725	32,454
SecP256r1MLKEM768	40,263	22,369	23,444
SecP384r1MLKEM1024	2,421	1,315	2,529

Signatures — ML-DSA

Algorithm	keygen	sign	verify
ML-DSA-44	19,597	3,204	17,188
ML-DSA-65	11,093	2,022	11,434
ML-DSA-87	7,970	1,733	7,333

TLS 1.3 handshake (`perftools`, in memory, single core, ML-DSA-44 certificate)

Provider	Key exchange	μs / handshake	Handshakes / sec
fips	X25519MLKEM768	654.5	1,528
default	X25519MLKEM768	633.0	1,580
default	SecP256r1MLKEM768	659.3	1,517
default	X25519	586.5	1,705

Provenance

file	rows	sha256
`primitives.csv`	108	`23643831068a0aee…`
`handshake.csv`	24	`0cc614a62658c495…`
`environment.csv`	7	`222289f92146a434…`

Notes

- QudoSSL runs in **FIPS mode**; the FIPS tables are the primary figures. Under the FIPS provider **key generation is materially slower** (mandatory pairwise-consistency test); encapsulation, decapsulation, signing and verification match the default-provider rates.
- Wall-clock and **indicative**: CPU frequency, core type and scheduling all affect it. The handshake harness runs in memory with no network, so its rate is an upper bound, not a deployment throughput.